

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement: Digital Gift Card & Loyalty Point Exchange

1. Introduction & Scenario Review

The assigned problem is Digital Gift Card & Loyalty Point Exchange, a digital wallet platform that allows an Account Holder to aggregate loyalty points from different Merchant Partners, convert those points into universal exchange tokens using dynamic conversion rates, and redeem the tokens for digital gift cards. The system also includes anti-fraud voucher locking to prevent duplicate or unauthorized voucher redemption.

The main actors identified in the scenario are the Account Holder and the Merchant Partner, supported by an Authentication & Security System.

2. Elicit & Draft Requirements

2.1 Brainstorm – Possible Functions

- View loyalty points from different Merchant Partners.
- Validate merchant loyalty points.
- View the current point-to-token conversion rate.
- Convert loyalty points into universal exchange tokens.
- View available digital gift cards.
- Redeem a digital gift card using exchange tokens.
- Generate a unique digital voucher.
- Lock vouchers to prevent duplicate redemption.
- Prevent expired or duplicate voucher redemption.
- Authenticate Account Holders before accessing their points and vouchers.

2.2 Brainstorm – Possible Constraints

- Point conversion and gift-card redemption should complete within 250 ms.
- The system should maintain zero ledger balance drift.
- Each voucher must be single-use.
- Invalid or expired loyalty points must not be converted.
- Duplicate vouchers must not be redeemable.
- Only authorized Account Holders should access their loyalty points and vouchers.
- The system must use the current dynamic conversion rate when converting points.

2.3 Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall validate loyalty points, convert them to exchange tokens, and generate single-use vouchers.	High	Valid points generate a unique voucher; invalid/duplicate redemption fails.	Core exchange function.	
FR-002	Functional	The system shall display an Account Holder's loyalty points and token value.	High	Correct points and token value are displayed.	Shows available value.	

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-003	Functional	The system shall convert eligible loyalty points using the current dynamic rate.	High	Correct token value is credited after conversion.	Enables point conversion.	
FR-004	Functional	The system shall allow users to redeem gift cards using exchange tokens.	High	Gift card is issued and tokens deducted when balance is sufficient.	Enables gift-card redemption.	
FR-005	Functional	The system shall lock vouchers and prevent duplicate redemption.	High	A redeemed voucher cannot be redeemed again.	Prevents fraud.	
NFR-001	Nonfunctional	The system shall process conversion and redemption within 250 ms with zero ledger drift.	High	Processing ≤ 250 ms and ledger remains accurate.	Ensures speed and accuracy.	
NFR-002	Nonfunctional	The system shall restrict loyalty points, tokens, and vouchers to authorized users.	High	Unauthorized access attempts are rejected.	Protects user assets.	

3. Peer Critique & Revision

The requirements table above was reviewed for clarity, testability, and relevance, and revised accordingly before finalizing actors and use cases. Comments from the peer review pass are captured in the "Comments" column of the table above.

4. Actors & Use Cases

4.1 Actors

Actor	Role
Account Holder	Views loyalty points, converts points, and redeems gift cards.
Merchant Partner	Provides and validates loyalty points.
Authentication & Security System	Authenticates users and supports secure voucher/transaction access.

4.2 Use Cases

Use Case ID	Use Case Name
UC-01	View Loyalty Points
UC-02	Convert Loyalty Points
UC-03	Generate Digital Voucher
UC-04	Redeem Gift Card
UC-05	Lock Voucher

5. UML Use-Case Diagram

Figure 1 below shows the use-case diagram for the Digital Gift Card & Loyalty Point Exchange system, including the system boundary, all actors, all use cases, and the «include» relationships between Redeem Gift Card, Convert Loyalty Points, and Lock Voucher.

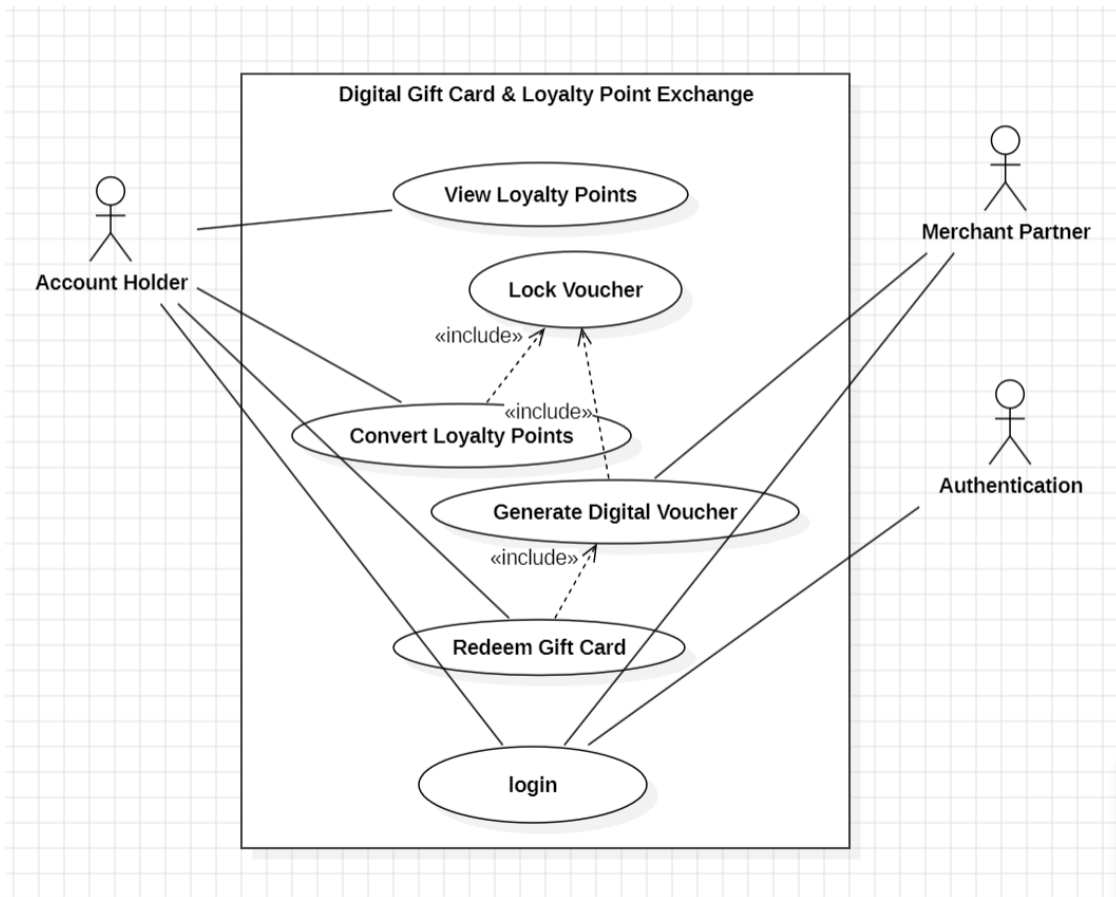


Figure 1. Use-case diagram – Digital Gift Card & Loyalty Point Exchange

6. Use-Case Flow: Redeem Digital Gift Card

6.1 Goal

Allow an Account Holder to redeem a digital gift card using available universal exchange tokens.

6.2 Use Case

UC-04 – Redeem Gift Card

6.3 Actors

- Account Holder
- Authentication & Security System

6.4 Main Success Scenario

1. Account Holder selects Redeem Gift Card.
2. System displays available digital gift cards and their token values.
3. Account Holder selects a gift card.
4. System verifies the Account Holder's token balance.
5. System deducts the required tokens.
6. System generates a unique single-use voucher.

7. System locks the voucher to prevent duplicate redemption.
8. System displays the digital gift card voucher to the Account Holder.
9. Use case ends successfully.

6.5 Alternate Flow – 4a. Insufficient Token Balance

- 4a1. System detects that the Account Holder has insufficient tokens.
- 4a2. System displays an Insufficient Balance message.
- 4a3. No tokens are deducted.
- 4a4. Account Holder can select another gift card or exit the redemption process.
- 4a5. Use case ends without completing the redemption.